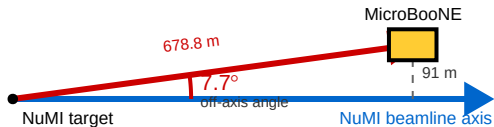


(a) beamline geometry

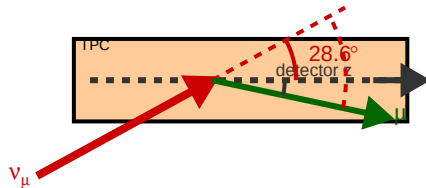
where the detector sits relative to the NuMI beam



This angle says nothing about how the TPC is oriented.

(b) inside the detector

the same ν , in TPC coordinates



the same μ is assigned

$\theta_\mu = 12^\circ$ from detector z

$\theta_\mu = 40.6^\circ$ from the ν

the generators mean the second

(c) the consequence

true $\text{CC}\mu\pi$ signal, FHC

64.6% of events change bin

